

Type-Aware Failure Prediction for Chart and Diagram Reasoning in Vision-Language Models

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1 Introduction and Motivation

Vision-language models (VLMs) are increasingly deployed to read charts, plots and diagrams where the output feeds a decision: financial summaries, dashboard question answering, figure captioning. When such a model answers incorrectly, it can be wrong in two mechanically different ways.

It may have **misread the figure** - picked the wrong bar, misjudged an axis tick, traced the wrong edge. Here the correct answer was present in the image and the model might recover if forced to look more carefully. Or it may have **fabricated**, stating a value or label appearing nowhere in the figure. Here looking again will not help, and the answer should be withheld entirely.

Current reliability tooling collapses these into a single verdict. A system that abstains on everything it distrusts discards recoverable answers; one that re-queries everything wastes compute on answers never grounded to begin with. This proposal asks whether the distinction is visible in the model’s internal state before generation, and whether knowing it is worth anything.

Contributions. A test of whether VLM failure *types* on structured images are linearly separable from pre-generation activations; a control, the probe-surface gap, separating a mechanistic signal from an artefact of our own labels; a cross-intervention matrix establishing whether different failure types require different repairs, independently of whether any probe succeeds; and an evaluation of a type-aware policy against a binary selective-prediction baseline and a label-aware oracle.

2 Background and Related Work

2.1 Pre-generation failure detection

Kogilathota et al. (2026) establish that hallucination risk in VLMs is readable from internal rep-

resentations in a single forward pass, without decoding, reaching up to 0.93 AUROC across eight models. Two of their findings shape our design: the most informative representation family varies by architecture, with Qwen2.5-VL-7B best served by visual-only features at roughly 0.79 AUROC where most models rely on late query-token states.

Zhang et al. (2026) localise the signal further, isolating hallucination-relevant attention heads through a variational information bottleneck and using probe gradients to intervene at inference time. Both works address binary object hallucination on natural photographs. Neither asks whether distinct kinds of failure are internally distinguishable.

2.2 Evidence that failure types differ mechanistically

Liu et al. (2026) apply activation patching across five VLMs and identify a visual grounding pathway supporting correct predictions alongside a separate hallucination pathway driving erroneous ones, the former at mid-to-late depths and the latter concentrated at early layers and network boundaries. Critically for us, the identified circuit transfers selectively to relational but not attribute hallucination.

That selectivity is the closest existing evidence that hallucination is not one internal phenomenon, and is the empirical basis for our hypothesis. It also motivates probing across the full depth of the network, since the two pathways do not occupy the same layers.

2.3 Chart and diagram reasoning

ChartQA (Masry et al., 2022) remains the standard benchmark for chart question answering with visual and logical reasoning. Rani et al. (2025) contribute chart, question, reasoning-step and attribution annotations, identifying which regions justify a model’s answer. Iyengar et al. (2026) assemble diagram QA instances across six source datasets, annotating the minimal regions required to verify

each answer, and find that correct answers do not guarantee a model grounded its reasoning in the supporting regions - models may instead rely on textual correlations and dataset artefacts.

Both resources annotate *where a model should look*. Neither labels *how a model failed*, which is the supervision signal we need.

2.4 Error taxonomies and visual self-correction

Ashury-Tahan et al. (2026) argue that benchmark accuracy reports whether a model fails but not why, and build an LLM-judge pipeline for taxonomy construction across 35 datasets and 83 models. Their setting is text-only LLMs; we borrow the methodology, not the taxonomy.

Li et al. (2026) propose Visual Self-Refine, in which a chart-parsing model emits pixel-level localisations, visualises them, and feeds them back to itself to correct its own perception errors. Making a model re-examine a figure is therefore established; what is untested is whether the *appropriate* repair depends on which failure occurred.

2.5 Probe validity

Two results define our standard of evidence. Sahoo et al. (2026) report probes separating reasoning types at 100% accuracy with well-separated geometry, then show the separation was driven entirely by format confounds, collapsing to chance once source identity, option count and response length were residualised out. Yuan et al. (2026) report the healthy contrast - a hidden-state probe at 0.95 AUROC against a text-surface classifier at 0.59 - but find that four interventions built on that signal failed: the signal was diagnostic, not causal.

Two lessons: a probe result is uninterpretable without a surface baseline, and detection does not imply a usable intervention.

2.6 Positioning

Pre-generation detection is binary and targets natural images; mechanistic type-selectivity has been shown for object hallucination, not structured-image reasoning; chart resources annotate evidence regions, not failure types; and visual self-correction is applied uniformly. Our contribution sits in the intersection: failure-type prediction on structured images, and whether that type should determine the repair.

3 Hypothesis

Type Separability Hypothesis. The pre-generation internal state preceding a structural misreading is linearly distinguishable from the state preceding a fabrication, because the former is a precision-limited perceptual failure while the latter reflects generation ungrounded in the image.

We are testing this, not assuming it. Charts pass through the same vision encoder as photographs, and misreading a bar height may be a higher-precision instance of the same perceptual failure rather than a distinct mechanism. Our experimental design is arranged so that a collapse of the two classes is still a reportable finding rather than a dead end (Section 6).

4 Methodology

4.1 Models and activation extraction

We use two frozen open VLMs, Qwen2.5-VL-7B and LLaVA-NeXT, with ChartGemma as a stretch goal. No weights are updated at any point; the only trained components are small probes and a controller.

For each (figure, question) pair we cache activations at a fixed set of layers spanning early, middle and late depths, at both vision-token and query-token positions, following HALP’s taxonomy of representation families and the depth-dependence reported by Liu et al. (2026). Storage rather than compute is the binding constraint, so we subsample layers rather than caching all of them.

4.2 Probes

We train three probes on cached activations, all logistic regressions with L_2 regularisation: a **binary** probe (correct vs. incorrect), which replicates HALP on structured images and serves as our sanity baseline - if it fails, nothing downstream is interpretable - and two one-vs-rest probes, **structural** and **fabrication**. We report per-layer AUROC for each, selecting layers on validation data only.

4.3 Cross-intervention matrix

This experiment does not use the probes at all. Taking only incorrect answers with *ground-truth* failure labels, we apply every repair to every failure type and record recovery: I_0 (no intervention, giving the baseline rate), I_{crop} (re-query at higher resolution on a question-conditioned crop), I_{verify} (re-query with a prompt demanding the model name the figure region supporting its answer) and

I_{abstain} (withhold the answer). Each cell reports $P(\text{correct} \mid \text{failure type } i, \text{intervention } j)$, with abstention scored as risk avoided rather than accuracy gained. The design question is whether this matrix has diagonal structure. If cropping rescues fabrications as often as misreadings, the taxonomy earns nothing however well the probes perform, and we would rather learn that directly than infer it from probe accuracy.

We note that the crop must be conditioned on the question rather than on the model’s own attention: cropping to where a model attended when it misread would reproduce the error.

4.4 Type-aware controller

Given a matrix with structure, we train a controller mapping cached activations to an intervention choice, and compare three settings: an **oracle** that sees the true failure type (the ceiling on what type-awareness can buy), the **learned** controller reading only activations, and a **binary** policy that abstains on everything the binary probe distrusts.

5 Data

5.1 Naturalistic arm

We run both VLMs over ChartQA (if time permits, on other structural datasets as well), collect incorrect answers, and label each with an LLM judge given the figure, question, gold answer and model answer, following [Ashury-Tahan et al. \(2026\)](#). Roughly 200 items are independently annotated by two team members to measure agreement. We set $\kappa > 0.6$ as the acceptance threshold; below it we simplify category definitions, adjudicate disagreements, and re-run rather than proceeding on unreliable labels.

5.2 Synthetic arm

We programmatically generate bar charts and node-link diagrams with known ground truth. This arm exists for three reasons. Labels are correct by construction rather than inferred by a judge. We can build matched item pairs that hold visual precision fixed while varying relational depth, and vice versa, separating the two factors by design instead of post hoc. And we can control the class balance directly - fabrication errors are expected to be scarce in natural data, and the cross-intervention matrix of Section 4.3 needs adequate counts in every cell.

Splits are constructed so no figure appears in both train and test.

5.3 Taxonomy

We deliberately keep two error classes rather than a finer-grained scheme. Splitting further - into localisation, association and scale subtypes - would multiply the annotation burden, degrade judge agreement, and leave too few examples per class to support reliable probing.

6 Evaluation Plan

E1: Probe quality. Per-layer AUROC for all three probes, with the binary probe checked against HALP’s reported range as a sanity condition.

E2: Probe-surface gap. Our categories correlate with cheap surface properties: “value absent from the figure” is nearly equivalent to “answer string absent from the figure’s text.” A probe encoding only that would score well while learning nothing about mechanism. We therefore train a classifier over surface features alone - answer present in figure text, answer type, answer length, question template, figure type - and report the margin by which the activation probe exceeds it. Following [Sahoo et al. \(2026\)](#) we additionally report probe accuracy after residualising these features out. The gap, not the raw AUROC, is our evidence.

E3: Separability. Each type probe is trained on its own class and tested for cross-transfer. Strong cross-transfer indicates one signal wearing two labels.

E4: Cross-intervention matrix. The full failure-type $\times$ repair recovery matrix of Section 4.3, with bootstrap confidence intervals per cell.

E5: Utility. Risk-coverage curves and AURC for the plain VLM, binary selective prediction, the learned controller and the oracle, at matched coverage. An ablation isolates whether re-querying contributes anything beyond abstention.

Outcomes. E4 and E1–E3 are independent, which is what makes the project robust. If the probes separate the classes and the matrix has diagonal structure, we have a working type-aware policy. If the matrix has structure but the probes fail, we report that different failures need different repairs but cannot be anticipated from internal state - a negative result about predictability, not about the taxonomy. If the matrix is flat, we report that the distinction does not pay, alongside the binary detection results, which would still be the first

pre-generation detection numbers for structured images.

7 Timeline

Period	Work
Aug 28 – Sep 10	Ada setup; VLM runs on ChartQA
Sep 11 – Sep 21	Judge labels; κ check; caching
Sep 22 – Sep 30	Probes; surface control
Oct 1 – Oct 15	Cross-intervention matrix; oracle
Oct 16 – Oct 31	Controller; risk-coverage; write-up

Under five weeks separate acceptance from the mid-submission, so the synthetic generator is built in parallel with the ChartQA runs. Sep 30 is our go/no-go checkpoint: by then we know whether the probes separate the classes and whether the surface baseline explains them away, which decides whether the second half pursues the controller or documents the negative result.

8 Risks and Mitigations

Low judge agreement is mitigated by the synthetic arm, whose labels need no judge; fabrication scarcity by asking about quantities outside a figure’s range. If the signal proves diagnostic but not causal - the failure mode Yuan et al. (2026) encountered - E4 still stands alone. Compute is not binding: 7B forward passes fit on Ada, probes take minutes on CPU, and the real costs are activation storage and judge API calls.

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